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Skill 26.1 Exercise 1

Refer to the movies DataFrame below.

	movie	production_budget	domestic_gross	worldwide_gross	mpaa_rating	genre
0	Evan Almighty	175000000.0	100289690.0	174131329.0	PG	Comedy
1	Waterworld	175000000.0	88246220.0	264246220.0	PG-13	Action
2	King Arthur: Legend of the Sword	175000000.0	39175066.0	139950708.0	PG-13	Adventure
3	47 Ronin	175000000.0	38362475.0	151716815.0	PG-13	Action
4	Jurassic World: Fallen Kingdom	170000000.0	416769345.0	1304866322.0	PG-13	Action

Suppose we want to compare the *production_budgets* for PG and R movies. Create two lists: one that has the *production_budget* for PG movies and another that has the *production_budget* for R movies.

```
budget_PG = movies[movies['mpaa_rating'] == 'PG']['production_budget']
budget_R = movies[movies['mpaa_rating'] == 'R']['production_budget']
```

Calculate the mean and median difference.

```
PG_mean = budget_PG.mean()
R_mean = budget_R.mean()
mean_diff = PG_mean - R_mean

PG_median = budget_PG.median()
R_median = budget_R.median()
median_diff = PG_median - R_median
```

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Skill 26.2 Exercise 1

Refer to the movies DataFrame below.

	movie	production_budget	domestic_gross	worldwide_gross	mpaa_rating	genre
0	Evan Almighty	175000000.0	100289690.0	174131329.0	PG	Comedy
1	Waterworld	175000000.0	88246220.0	264246220.0	PG-13	Action
2	King Arthur: Legend of the Sword	175000000.0	NaN	139950708.0	PG-13	Adventure
3	47 Ronin	NaN	38362475.0	151716815.0	PG-13	Action
4	Jurassic World: Fallen Kingdom	170000000.0	416769345.0	NaN	NaN	Action

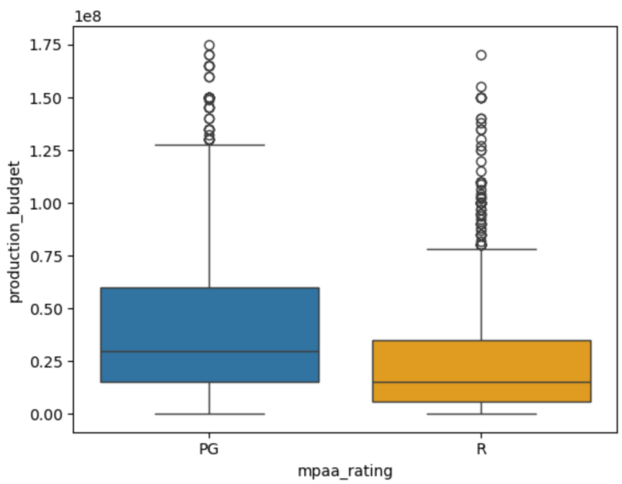
Filter the dataframe for the movies with an *mpaa_rating* of R and PG only. Store the results in a new dataframe, *PG_R*

```
PG_R = movies[(movies['mpaa_rating'] == 'PG') | (movies['mpaa_rating'] == 'R')]
```

Complete the spaces below to make side by side box plots for R and PG movies.

```
sns.boxplot(data =    PG_R              ,
            x =    'mpaa_rating'        ,
            y =     'production_budget' ,
            hue =  'mpaa_rating'        ,
            legend =  True              )
plt.show()
```

Below are the actual box plots for PG and R movies,



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Based on the box plots, does movie rating appear to be associated with budget — that is, do PG and R movies have different budget distributions?
Yes, movie rating appears to be associated with budget. PG movies tend to have higher median budgets and greater variability compared to R movies, suggesting that budget distributions differ by rating.

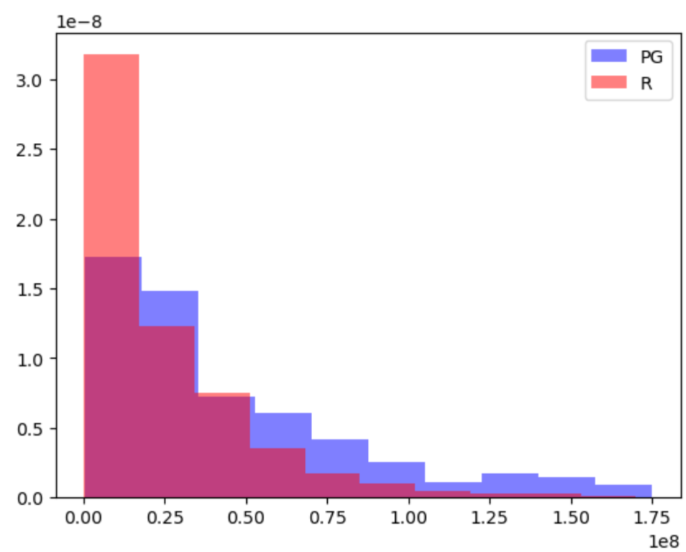
Skill 26.3 Exercise 1

Refer to the lists previously created in *Skill 26.1 Exercise 1*. One has the *production_budget* for PG movies and the other has the *production_budget* for R movies. Complete the spaces below to make overlapping histograms.

```
plt.hist(          , color="blue", label=      , density=       , alpha=        )
plt.hist(          , color="red", label=      , density=       , alpha=        )
plt.legend()
plt.show()
```

```
plt.hist(budget_PG , color="blue", label="PG", density=True, alpha=0.5)
plt.hist(budget_R , color="red", label="R", density=True, alpha=0.5)
```

Below are the actual overlapping histograms for PG and R movie production budgets,



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Based on the overlapping histograms, does production budget appear to be associated with movie rating? Support your answer with evidence from the graph.
The overlapping histograms show that **both PG and R movies are right-skewed**, with most budgets on the lower end. However, **PG movies appear more likely to have higher budgets**, while **R movies are more concentrated in lower budget** ranges. This suggests that production budget is associated with movie rating, although there is significant overlap between the two distributions.

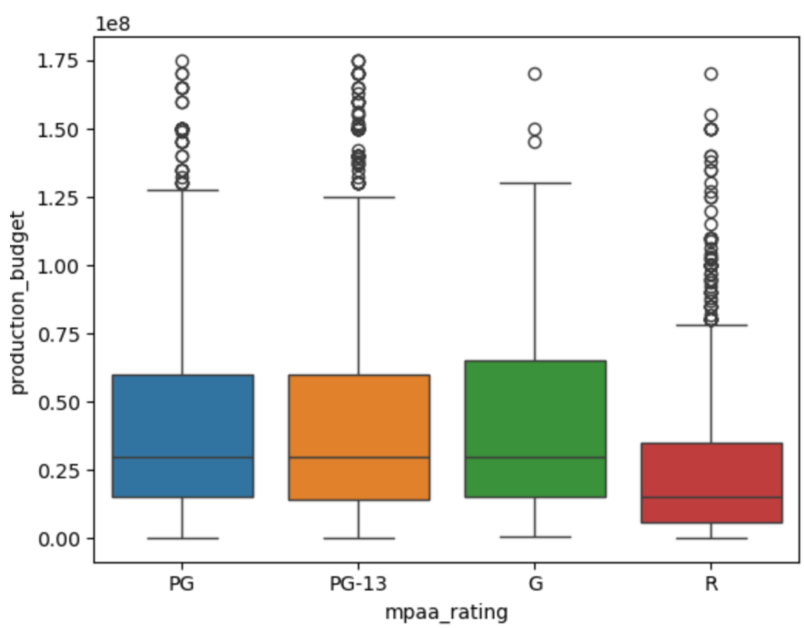
Skill 26.4 Exercise 1

Refer to the movies dataframe from above

Complete the spaces below to make side by side box plots for all *mpaa_ratings*

```
sns.boxplot(data =   movies             ,
            x =   'mpaa_rating'         ,
            y =   'production_budget'   ,
            hue =  'mpaa_rating'        ,
            legend =  False             )
plt.show()
```

Below are the actual box plots for all *mpaa_ratings*



Based on the box plots, does movie rating appear to be associated with budget?

The box plots suggest that production budget is associated with movie rating. R-rated movies tend to have lower median budgets and less variability, while G, PG, and PG-13 movies generally have higher median budgets and more high-budget outliers. Although there is overlap among the groups, the difference between R and the other categories appears noticeable.

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